Anish Natekar

TCS Research (TRDDC), Pune

Graduate Year **B.Tech**, **Computer Science and Engineering**
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Education

B.Tech, Computer Science and Engineering, Indian Institute of Technology Goa	CGPA: **8.35/10**	2020 – 2024
Class 12, ISC (ICSE), Hiranandani Foundation School, Powai	Aggregate: **86.8** %	2018 – 2020

Experience

Research Engineer, TCS Research — Generative AI — Aug 2024 – May 2026
- Architected multi-agent LLM systems for multimedia analysis using DSPy-based structured reasoning pipelines.
- Designed hybrid RAG systems integrating vector search, MCP-based tool calling, and evaluation feedback loops.
- Built automated prompt optimization and benchmarking frameworks to improve reasoning reliability.
- Conducted LoRA/QLoRA fine-tuning experiments on domain-specific datasets.

R&D Intern, Siemens EDA (Mentor Graphics) — Jul 2023 – Dec 2023
- Contributed to compiler optimization stage (vopt) in Questa HDL simulator.
- Enhanced X-value propagation logic and built regression tests in large-scale C codebase.
- Worked in Linux (CentOS) environment using Perforce Helix.

Publications

MemGrad: Memory-Guided Optimization of Agentic Software Development via Abstracted Textual Gradients — ICLRW 2026 — First Author
- Proposed memory-guided optimization framework converting multi-trajectory agent feedback into structured textual gradients.
- Designed retrospective–prospective memory architecture for improving multi-agent reasoning stability.
- Demonstrated performance gains in AgileCoder without model fine-tuning via automated prompt updating.

Democratic ICAI: Debating Our Way to Steering Principles from Preferences — ICLRW 2026 — Second Author
- Built an LLM decision-tree creation pipeline from human preference reasoning to determine when and what to query LLMs for alignment.

HyP-ECA: Attention for Aerial Tree Crown Delineation and Species Classification — CVIP 2024 — First Author
- Developed UAV-based aerial tree species detection pipeline.
- Proposed HyP-ECA attention integrated with YOLOv8 for improved small-object detection.
- Curated MTAD dataset and validated system via real UAV deployment and SITL simulation.

UAV Based Aerial Object Imagery for Animal Ecology — IEEE ETAAV 2025 — Second Author
- Contributed to WildYOLO (YOLO + Efficient Channel Attention) for wildlife monitoring.
- Evaluated on WAID dataset (mAP@50: 94.2%) and tested and deployed over Raspberry Pi-onboard UAV integrated with pixhawk Flight Controller.

Sitar V2: Parallel Simulation of Networked Synchronous Discrete-Event Systems — Winter Simulation Conference 2024 — Co-Author
- Modernized Sitar V2 framework (Ubuntu 20.04, Python 3).
- Developed optimized SimPy models and conducted performance analysis.

Projects

Gallery Search — On Device AI powered IOS/Android APP to search, organize, and cluster your Photo Gallery — Feb 2026 – Present
- Designed and Deployed Flutter based AI powered APP for IOS/Android that works completely offline on Device with help of Claude Code.
- Implemented Text, Img, Face based search on local Image Gallery with Ability to Cluster Images based on Semantic or Face similarity.
- Implemented Custom models, algorithms, and vector database for Image indexing, embedding, querying, and clustering on device.

Raspberry Pi PicoW Copter — Open-Source WiFi (IOT) Controlled Quadcopter (50+ GitHub Stars) — May 2023 – Jul 2023
- Designed full-stack quadcopter system: PCB, firmware, and Flutter-based control interface.
- Implemented wireless flight control using WiFI via UDP protocol on Raspberry Pi PicoW.
- Featured in international media including Tom's Hardware (Popular DIT Tech journalist site since 1996).

Skills

AI Systems & LLM Engineering:	Multi-Agent Systems, Prompt Optimization, RAG, ComfyUI, Model Benchmarking, LoRA/QLoRA Finetuning, Coding Agents, Agentic Harness, DSPY, Langchain, Autogen, MCP, ReAct Agents.
ML Frameworks & Tooling:	PyTorch, Hugging Face Transformers, PEFT, DSPy, LangChain, YOLO, OpenCV, ChromaDB, Vector Databases.
Programming & Infrastructure:	Python, C/C++, SQL, Linux, Docker, Azure, REST APIs, Git, System Verilog, Arduino, Flutter/Android, IOT.
Relevant Coursework:	Machine Learning, Artificial Intelligence, Deep Learning, Operating Systems, Computer Networks.

Positions of Responsibility

Freelance Gen AI Engineer	Delivered custom AI (CV, NLP, Gen AI) automation solutions for international clients; earned $1000+ with 5-star rating.	2026 – Present
Head, E&R (Robotics) Club:	Built line follower Robots, RC planes, and Quadcopter/Drones for Aerial Survey and sub 100g drones for swarm and educational applications.	(2022 – 2023)
Student Representative 2020:	Represented Batch of 2020 in front of the UGC (Undergraduate Committee) of IIT Goa	(2020 – 2024)

Achievements

- JEE Advanced AIR **3858**; JEE Mains **99.2 percentile**. — 2020
- 1st Place — IIT Goa Coding Contest (Game of Codes); 3rd Place — IIT Goa IIC Virtual Lab Hackathon. — 2023, 2021